One-sheet: Schrödinger from the bare CAL at $\alpha \to 1^-$

Schrödinger evolution from the bare Causal Accumulation Law at $\alpha \to 1^-$: feedback on the accumulated past, unitarity in the limit, and carrier entanglement as time dilation. S. Leizerman, LIOR Research Initiative, November 13, 2026. Script: `cal_edge_transducer.py` (numpy only, about 15 s).

Inputs.

- The CAL as written: $T = \alpha J - (1-\alpha)\int k\,\Pi P \Phi\, J$.
- The free, flat sector: $\Pi P \Phi = 1$, an order-α power-law memory, and a spatial profile $\hat{g}(k)$.
- The phase belongs to the source: order-α accumulation turns $e^{iS/\hbar}$ by α quarter turns.
- The update rule $J_{n+1} = T_n$, a feedback loop on the accumulated past.
- The current's past rotated one way.

Not used: a phase prefactor, a coupling constant, a time unit, a double scaling, or $T = 0$ as a law. Time is counted in radians of the carrier's own rotation.

Result	Statement	Check
Rotation	Below 1 the past pulls back, and the current turns by $\sqrt{1-\alpha}$ per update. That rotation is the i of Schrödinger.	ratio $0.9897 \to 1.0000$ for $1-\alpha = 10^{-2} \ldots 10^{-7}$
Dispersion and mass	The rotation slows as $\sqrt{\hat{g}(k)}$, so $\hbar/2m = \langle r^2 \rangle/12$ in rest-rotation units, set by the kernel width.	ratio 1.0000 to 0.990
Unitarity in the limit	For $\alpha < 1$ the evolution is a contraction, $\exp[-(i+\sqrt{1-\alpha})(\hbar/2m)k^2 t - \frac{\pi}{4}(1-\alpha)t]$. It is unitary as $\alpha \to 1^-$.	decay per radian $\div(1-\alpha)$: $0.901 \to 0.787$ ($\pi/4 = 0.785$)
Wavepacket	Converges to Schrödinger down to the k^4 floor, which halves with $\langle r^2 \rangle$.	1.46×10^{-2} vs floor 1.26×10^{-2}
Orientation	A one-way past keeps one direction. The whole past is needed: a two-cycle history leaks 70–97%.	leakage $\sim 10^{-8}$
Born needs one orientation	With a one-way past, the closure and readout drift $\to 0$. With a zero past, the readout drift stays at 3.7×10^{-2}.	Table 5 of the paper
Feedback across $\alpha = 1$	Frozen at 1. Above 1: a causal runaway at $\sqrt{\alpha - 1}$, or an advanced bounded response (the two ALD horns). Braking plus fading settles it; braking alone cannot (windup).	growth 0.101 vs 0.1; settles for $b > 0.095$ at $\lambda = 0.1$
Normalizable memory	Every root lies strictly inside $\lvert z \rvert = 1$, so nothing sustains a rotation. The power law escapes because its total weight is infinite.	proof + companion-matrix roots
Loss	Loss is contraction from rank 2 to rank 1: absent in the free edge, present in interaction and readout, where the Landauer cost is paid.	—
Kept	Galilean edge; no interaction (one order-1 leg); one carrier is two factors with $C = 1/\gamma$; readout survives composition; $\Lambda^2 J = 0$.	readout checks 10^{-15}–10^{-16}

Changed from the earlier version.

- The prefactor, r_1, τ_c, the double scaling and $T = 0$ are removed.
- Unitarity is now exact only at the edge, not for every $\alpha < 1$.
- The mass formula is $\hbar/2m = \langle r^2 \rangle/12$ in rest radians.

Open.

- A rigorous remainder estimate for the asymptotics. It is formal now, and confirmed numerically to $1 - \alpha = 10^{-7}$.
- The equivalence condition, a kernel width of $\sqrt{6}$ Compton wavelengths. It is a forward pointer only.
- Interaction, which needs the return leg (order 2).